

CS-2614 Lab 4

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Name

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1.

D_0	D_1	S_0	E	Y
X	X	X	0	X
0	0	0	1	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	1	1

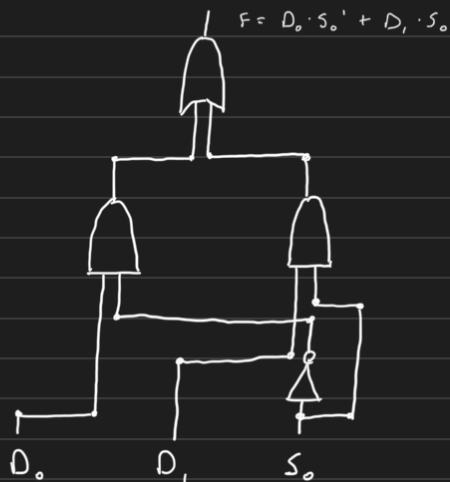
A \ BL	00	01	11	10
0			1	
1	1		1	1

$$A = D_0$$

$$B = D_1$$

$$C = S_0$$

$$F(A, B, C) = AC' + BC \equiv D_0 \cdot S_0' + D_1 \cdot S_0$$

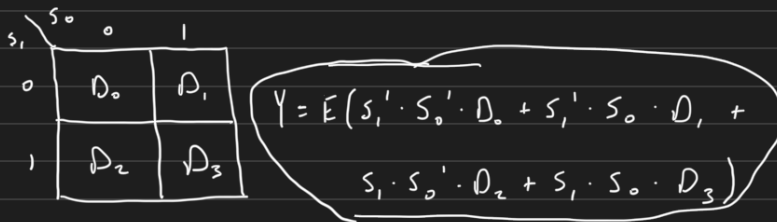


2.

a.

E	D ₀	D ₁	D ₂	D ₃	S ₀	S ₁	Y
1	0	X	X	X	0	0	0
1	1	X	X	X	0	0	1
1	X	0	X	X	1	0	0
1	X	1	X	X	1	0	1
1	X	X	0	X	0	1	0
1	X	X	1	X	0	1	1
1	X	X	X	0	1	1	0
1	X	X	X	1	1	1	1
0	X	X	X	X	X	X	X

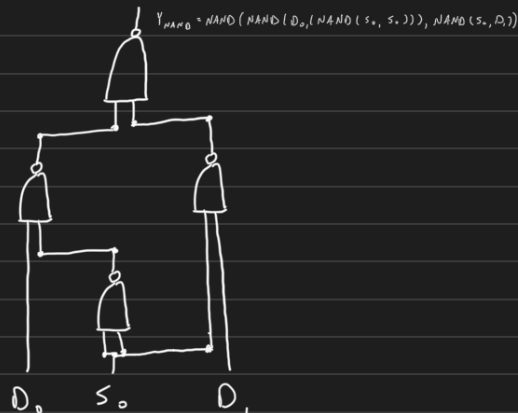
b.



c.

$$Y = D_0 \cdot S_0' + D_1 \cdot S_0$$

$$Y_{NAND} = NAND(NAND(D_0, (NAND(S_0, S_0))), NAND(S_0, D_1))$$



3.

First layer:

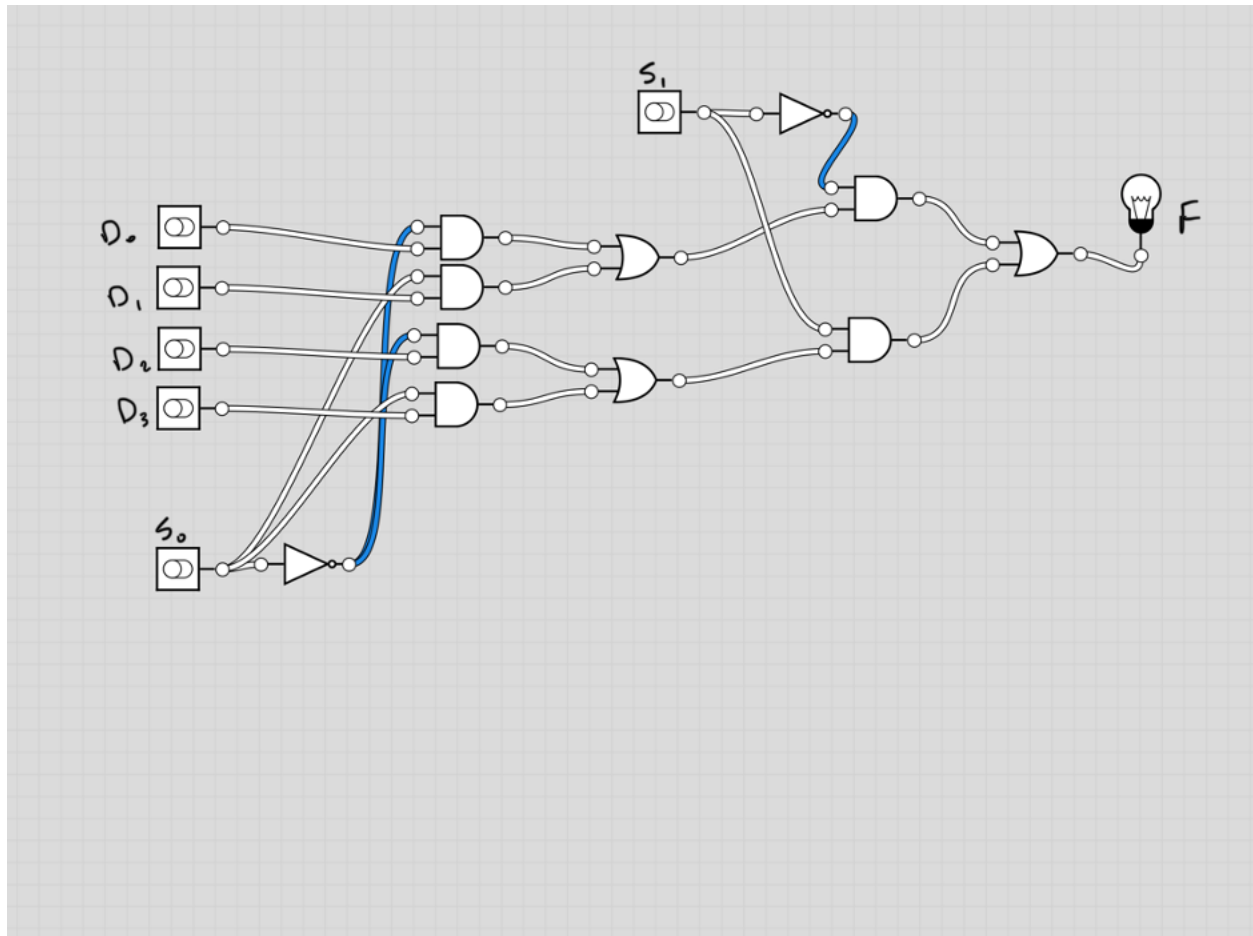
$$Y_1 = S_0' D_0 + S_0 D_1$$

$$Y_2 = S_0' D_2 + S_0 D_3$$

Second layer:

$$Y_3 = S_1' Y_1 + S_1 Y_2$$

Circuit for #3:



4.

